26900158 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours) REGULAR EXAMINATION October 2026

Fifth SEMESTER

Discipline Specific Elective (DSE) - MG5DSECSC304 - DESIGN PRINCIPLES OF USER COMPUTER INTERACTION

(2024 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Very Short Answer Questions

Answer all questions

Each question carries **2** marks

1. List two key advantages of capacitive touchscreens over resistive touchscreens. [K] / [CO1]
2. Summarize the primary limitation of predictive cognitive models. [U] / [CO2]
3. Define cart abandonment in e-commerce interaction design. [K] / [CO3]
4. Define Ubiquitous Computing (Ubicomp) as envisioned by Mark Weiser. [K] / [CO4]
5. List two interface design adaptations specifically needed for elderly users. [K] / [CO1]

[2x5 = 10]

Part B

Short Answer Questions

Answer any **6** questions

Each question carries **5** marks

6. Discuss the psychological impact of system response time and delay on user performance. [U] / [CO1]

7. Compare remote asynchronous usability testing with moderated in-person usability testing. [An] / [CO2]

8. Show how spatial audio can be employed in virtual meeting rooms to improve conversational clarity. [A] / [CO3]

9. Implement a dialog flow diagram that handles network timeout and retry semantics in a mobile app. [A] / [CO3]

10. Show how cognitive workload assessment using the NASA-TLX tool is conducted on an interactive system. [A] / [CO4]

11. Classify the various human sensory modalities according to their bandwidth and feedback speed in HCI. [U] / [CO1]

12. Implement a floor control protocol for a multi-speaker virtual seminar room. [A] / [CO4]

13. Implement Nielsen's 'Error prevention' heuristic in the design of a date-picker and flight booking form. [A] / [CO3]

[5x6 = 30]

Part C

Essay Questions

Answer any **2** questions

Each question carries **15** marks

14. Examine the information architecture, navigation paradigms, and interaction design patterns of large-scale social media platforms, analyzing their impact on user engagement and cognitive overload. [An] / [CO2]

15. Show how affordances, signifiers, mapping, feedback, and constraints are systematically implemented to design an intuitive multi-functional kitchen appliance control panel. [A] / [CO3]

16. Illustrate the complete interaction design and spatial user interface (3D UI) architecture for an Augmented Reality (AR) surgical training and anatomical visualization system. [A] / [CO4]

[15x2 = 30]